



# Device and circuit-level performance comparison of GAA nanosheet FET with varied geometrical parameters

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## ABSTRACT

In this paper, DC and analog/RF figures of merit (FOMs) for different geometrical variations of the Gate all around (GAA) Nanosheet FET (NSFET) are computationally examined. For each nanosheet, the thickness ( $N_T$ ) varied from 5 nm to 9 nm and width ( $N_W$ ) varied from 10 nm to 50 nm to analyze the performance variations with the device's geometry. It is observed that DC metrics like switching ratio, drain induced barrier lowering (DIBL), and subthreshold swing (SS) degrade with increment in geometry of NS. Moreover, the increment in analog/RF FOMs like transconductance ( $g_m$ ), output conductance ( $g_{ds}$ ), cutoff frequency ( $f_T$ ), transconductance frequency product (TFP), and gain bandwidth product (GBW) is observed as the width and thickness are increase towards 50 nm and 9 nm, respectively. However, intrinsic gain, GFP, and GTFP deteriorate as the physical dimensions of the NSFET increase. On top of that, inverter and ring oscillator (RO) circuits are designed and the performance is demonstrated by varying width of the nanosheet. The circuit analysis is performed in Cadence Virtuoso tool by using look-up table based Verilog-A model. A decrement of 17.6% in the delay of the inverter is noticed when the width is increased from 10 nm to 50 nm. Also, it is observed that the static current of inverter is less than pA, which ensures less static power dissipation. Furthermore, an increment of 46.5% in  $f_{osc}$  of 3-stage RO is observed as width increased to 50 nm. According to the various results analyses, NSFET is a promising device for high performance and high frequency analog/RF applications for sub-7-nm technology nodes.

## 1. Introduction

The semiconductor devices have been significantly improved since 1992 when the ITRS started to issue the roadmaps for semiconductor technology. These roadmaps define the industry trends for their improved productivity and performance. Also, they proclaim the ways to continue Moore's law. The miniaturization of the transistors has been following Moore's law and remarkably leads to high throughput, lower energy consumption, high speed, and higher integration of the transistors. By downscaling the physical dimensions, the device can be operated at high speed at higher operating frequencies and consumes less power. According to Moore's Law, there is a two-fold increment in the transistors count on a chip for every one and half years. Moreover, downscaling the device leads to easy portability and cost reduction [1–4]. In addition to the gate length, device dimensions are also reduced for each technological node [5]. In conventional metal oxide semiconductors because of aggressive scaling tends to various short channel effects (SCEs). The fields from the source/drain start influencing the region under the gate, which invites the detrimental SCEs in

conventional FETs. In transistors, SCEs lead to an increase in energy loss and performance deterioration. In order to have control over power consumption, in addition to the device dimensions, the supply voltage ( $V_{DD}$ ) also needs to be scaled. To maintain high drive currents, fewer internal electric fields and good performance in short channel transistors, the threshold voltage ( $V_{th}$ ) also needs to be scaled. However, the decrement in threshold voltage leads to a substantial increment of sub-threshold current leakage. The leakage current ( $I_{OFF}$ ) is affected by various performance metrics like threshold voltage, doping profile, device's physical dimensions, the junction depth of source and drain, the thickness of gate oxide, and  $V_{DD}$ . Some of the short channel effects are threshold voltage roll-off, degradation of subthreshold slope (SS), drain induced barrier lowering (DIBL), etc [6]. While scaling down the transistor, to keep control of SCEs, the gate oxide thickness also needs to be scaled in proportional to the channel length. However, the reduced thickness of gate oxide leads to a considerable amount of current flow from the gate and destroys the advantageous concept of the infinite input impedance of the MOSFETs. To circumvent these SCEs, various device engineering and new structures with multiple gates are proposed.

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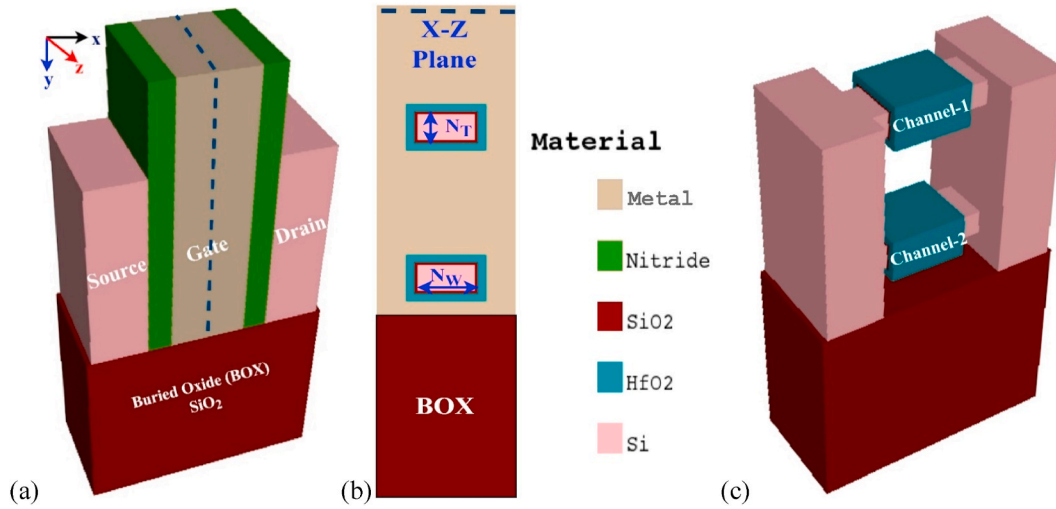


Fig. 1. (a) Three-dimensional view of NSFET with Nitride spacer, (b) Cross sectional view of NSFET with  $N_T$  and  $N_W$  and (c) Three-dimensional view of NSFET with nanosheet channels.

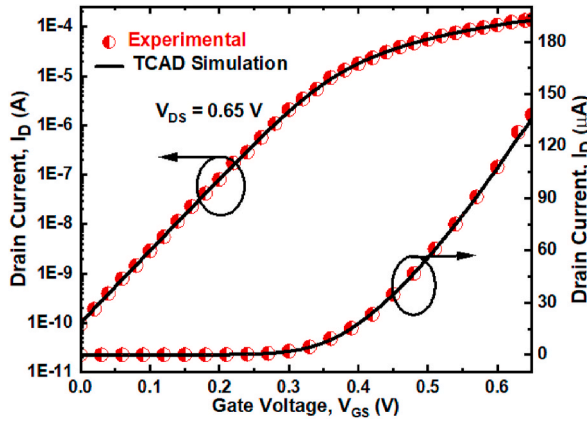


Fig. 2. Validation of simulation models with that of experimental data [27].

By stacking a ferroelectric layer in the gate stack, the Negative Capacitance FET (NCFET) can be realized which offers steeper switching slope. The detailed analysis of NCFET and its mixed signal applications are discussed in Refs. [7–11]. Also, advanced multiple gate structures like FinFET and Nanosheet/Nanowire structures are proposed to mitigate SCEs [12–20].

Double gate MOSFETs contain two gates, in which gates are present at the bottom and top of the device. By maintaining the two gates, the channel's electrostatic integrity increases and offers better short channel performance. However, in double gate MOSFETs, the misalignment problem of the top gate and bottom gate is present, which causes degradation in on current [21]. In trigate and FinFET devices, the gates surround the channel in the left, top and right directions and are possible candidates to replace the orthodox MOSFETs in the semiconductor industry. Moreover, FinFET structures are more immune to SCEs and offer fewer DIBL and SS values for sub-22-nm technology nodes. The semiconductor companies like TSMC and INTEL have adopted FinFETs into their significant portion of IC fabrication since 2011. However, for further scaling of FinFETs, taller and thinner fins are required to get more drive current, which increases the process complication of the device. Moreover, maintaining the taller fins on a narrow base may lead to fracture and malfunction of the circuit [22].

To circumvent these problems, researchers came up with the solution of GAA configurations. Since the channel in GAA structure is controlled in all directions by the gate and hence improves the sub threshold

performance. Nanosheet (NS) and nanowire (NW) are examples of GAA structures. On the other hand, it is reported that, due to the surface roughness factor, the nanowire performance is deteriorated [23]. The GAA nanosheet structures are FinFET successors for sub-7-nm technology and beyond, and have emerged as viable challengers for traditional FinFETs since they offer greater control over the channel region. Robust leakage control and strong current driving capacity are two advantages of nanosheet architectures. Due to the tight electrostatic control over the channel region, NSFETs are resistant to short channel effects [24]. S. D Kim reported that NSFETs exhibited greater electrostatic performance on the channel compared to GAA nanowire and FinFET architectures [25]. By making minimal changes to the fabrication flow of FinFET, NSFET can be fabricated as shown in Ref. [26]. Also, it is shown that by using NSFETs instead of FinFETs, 30% higher effective width ( $W_{eff}$ ) can be produced on the same footprint. Another remarkable observation is that single stack nanosheets have been shown to have superior intrinsic performance than FinFETs or stacked nanowires for a given active width due to lower parasitic capacitance. In a given footprint, nanosheets offer a higher effective width making them good at driving capacitive loads [27,28]. Moreover, nanosheet width doesn't affect by fin pitch which provides greater flexibility in maintaining appropriate  $W_{eff}$ . Usage of more stacked nanosheets to maximize the drive current is a better option, but the larger height of the device exacerbates parasitic capacitances [29].

In this paper, the performance of two stack GAA NSFET at 5 nm technology node is designed towards ITRS projections, and various DC and analog/RF design metrics are discussed. Further, high- $k$  dielectric material  $HfO_2$  is involved in the gate stack to reduce the gate leakage current [30]. Although some studies discussed the performance evaluation of NSFET with different geometry at the device level [33], the impact of NS width ( $N_W$ ) on the voltage transfer characteristics (VTCs) of CMOS inverter, propagation delay ( $\tau_p$ ), switching current ( $I_{sc}$ ) and frequency of oscillations ( $f_{osc}$ ) of 3 stage ring oscillator circuits still needs to be discussed. The efforts are made in this paper to evaluate these parameters in detail.

In order to evaluate the NSFET performance towards circuit applications, digital circuits like inverter and ring oscillator (RO) are designed and demonstrated their performance. The paper is organized as follows. The device physics and parameters are discussed in section 2 of the manuscript. The geometrical dependence on NSFET towards DC, and analog/RF performance is investigated in section 3. For all the results presented in section 3, n-type NSFET is considered. By using both n-type NSFET (N- NSFET) and p-type NSFET (P- NSFET), circuit level performance is explored in section 4. Further, section 5 concludes the paper.

**Table 1**  
Simulation parameters.

| Parameter                                                                                                   | Value     |
|-------------------------------------------------------------------------------------------------------------|-----------|
| Thickness of nanosheet (nm)                                                                                 | 5–9       |
| Width of nanosheet (nm)                                                                                     | 10–50     |
| Length of gate (nm)                                                                                         | 16        |
| Source/Drain doping ( $\text{cm}^{-3}$ )                                                                    | $10^{20}$ |
| Doping of the channel ( $\text{cm}^{-3}$ )                                                                  | $10^{15}$ |
| EOT ( $T_{\text{SiO}_2} + (\epsilon_{\text{SiO}_2}/\epsilon_{\text{HfO}_2}) \times T_{\text{HfO}_2}$ ) (nm) | 0.78      |
| Dielectric of Underlap                                                                                      | Nitride   |
| Pad length of Drain/Source (nm)                                                                             | 12        |
| Source/Drain underlap length (nm)                                                                           | 5         |
| Gate work function (eV)                                                                                     | 4.6       |
| Height of the gate (nm)                                                                                     | 60        |

## 2. NSFET device and simulation parameters

The three-dimensional diagram of GAA NSFET is depicted in Fig. 1 (a). 2-D cross sectional view in XZ plane is shown in Fig. 1 (b), where  $N_T$  and  $N_W$  are the thickness and width of each nanosheet respectively. All the simulations are performed through the Cogenda 3D TCAD simulator [31]. A gate length ( $L_G$ ) of 16 nm is considered for NSFET. A p-type doping concentration of  $10^{15} \text{ cm}^{-3}$  and n-type doping concentration of  $10^{20} \text{ cm}^{-3}$  for channel and source/drain, respectively used for simulation. To overcome the gate oxide tunneling a high-k gate stack with  $\text{HfO}_2$  of 1.5 nm and interfacial oxide of 0.5 nm, which forms an effective oxide thickness (EOT) of 0.78 nm is considered. The metal gate with the work function of 4.6 eV is maintained throughout the simulations. To overcome the direct tunneling a spacer distance of 5 nm is incorporated, which improves the sub-threshold performance [32].

The device is well calibrated using [27,33], and the  $I_D$ - $V_{GS}$  characteristics are depicted in Fig. 2. The physical models incorporated for simulation are considered from Ref. [33] and are as follows. Schenk's bandgap narrowing model is used to account for narrow bandgap effects caused by larger doping levels. The Shockley-Read-Hall (SRH) recombination method is used to assess the consequences of carrier generation and recombination. The Lombardi mobility model is used to account for scattering phenomena such as surface roughness. The Selberherr impact ionization model is used to calculate the electron-hole pair's generation rate. The simulation models are validated by using experimental data. Table 1 provides the values of device parameters that are included in the simulation.

## 3. Results and discussion

In this section, both DC and analog/RF FOMs are discussed for NSFET. To investigate the performance dependence on NSFET geometry, a lower and upper bound of 5 nm and 9 nm for  $N_T$ , 10 nm and 50 nm

for  $N_W$  are considered. Section 3.1 deals with DC performance dependence on geometry by considering various DC performance metrics. Analog/RF FOMs are discussed in section 3.2.

### 3.1. Geometry dependence on DC metrics of NSFET

In this section, the NSFET's DC metrics are evaluated with various thickness and width values and are discussed in detail. The major performance characteristics mentioned in this section include  $I_{ON}$ ,  $I_{OFF}$ , switching ratio ( $I_{ON}/I_{OFF}$ ), threshold voltage ( $V_{th}$ ), DIBL, and SS.

The dependence of electron mobility on the channel in ON state is depicted in Fig. 3(a). The device exhibits higher mobility in the channel area due to better gate electrostatics. Fig. 3(b) depicts the electric field distribution of NSFET. The device exhibits a higher electric field in the spacer region. However, the impact is lower on the channel. Fig. 3(c) depicts the potential distribution of NSFET at  $V_{DS} = V_{GS} = 0.7 \text{ V}$ .

The rise in drain current and parasitic capacitance is a trade-off when using high-k spacers to reduce circuit delay. Enhanced spacer designs will be required to improve circuit and device performance even more by lowering parasitic capacitances while maintaining a high drive current [34]. In our work, the nitride spacer with an optimized thickness of 5 nm is considered, which offers lower degradation of the  $I_{eff}/C_{eff}$  electrical performance [27]. The introduction of high-k spacers induces field coupling through the fringing effect and enhances switching ratios [35]. By using nitride spacer due to the more distance between channel and drain ensures reduced drain potential over the channel and

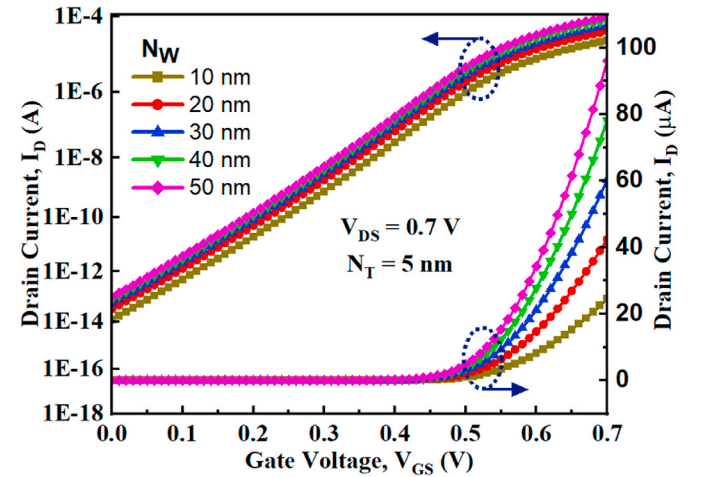


Fig. 4. Transfer characteristics of NSFET in linear and log scale for various  $N_W$  values.

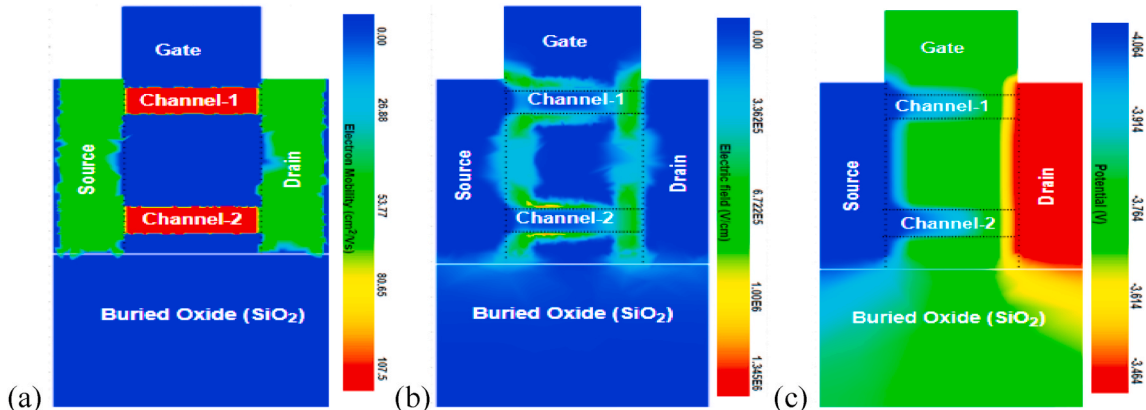
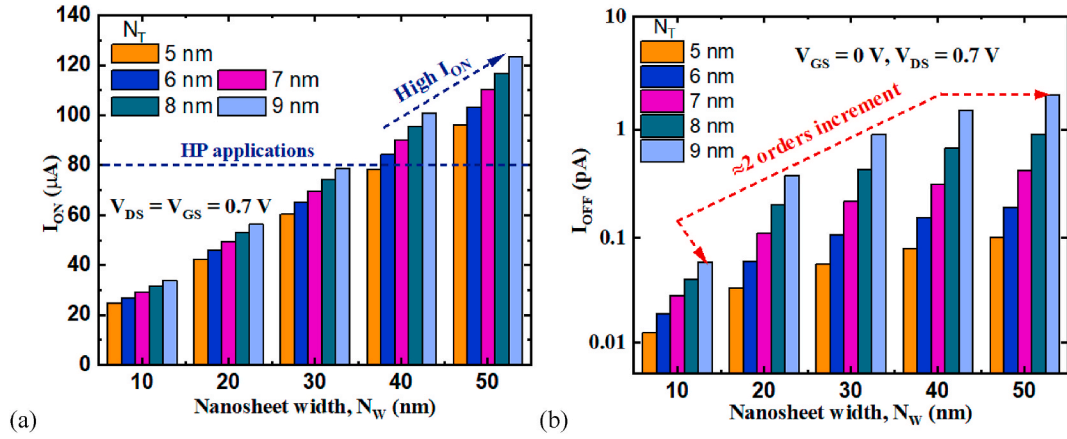
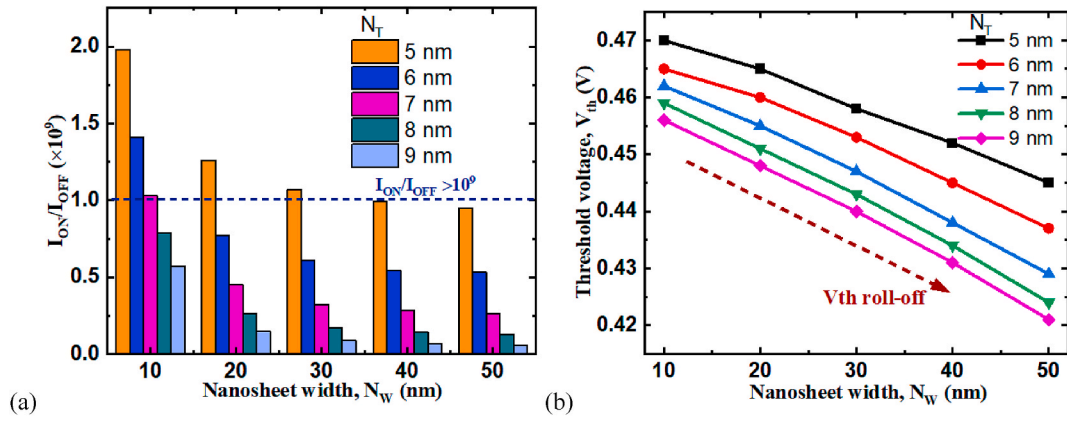
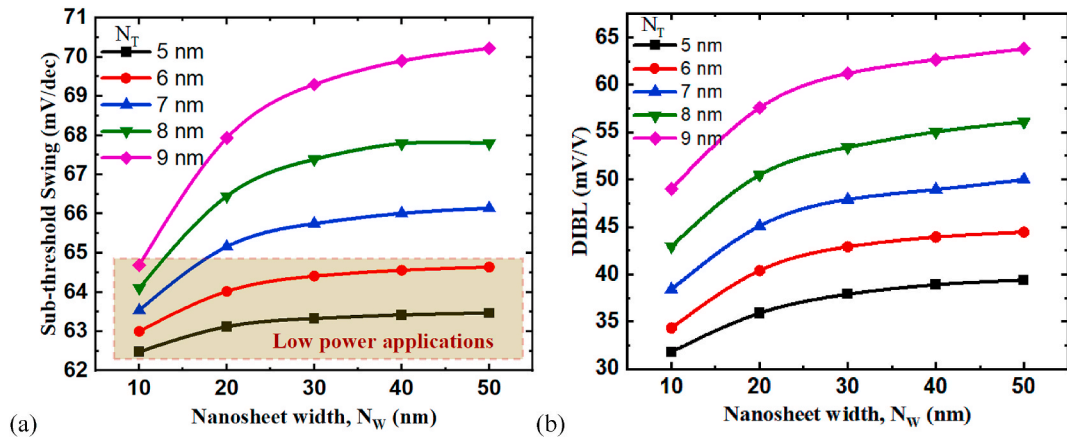


Fig. 3. (a) Electron mobility (b) Electric field distribution (c) Potential of NSFET.

Fig. 5. (a)  $I_{ON}$  and (b)  $I_{OFF}$  as a function of  $N_W$  for various  $N_T$  values.Fig. 6. (a) Switching ratio and (b) Threshold voltage ( $V_{th}$ ) variations as a function of  $N_W$  for various  $N_T$  values.Fig. 7. (a) SS and (b) DIBL as a function of  $N_W$  for various  $N_T$  values.

minimizes SCEs, as shown in Fig. 3(c).

Fig. 4 shows the transfer characteristics NSFET with various widths at constant  $L_G$  of 16 nm and  $N_T$  of 5 nm. The flow of current in the nanosheet is dependent on the width of the nanosheet. In order to evaluate the impact of  $I_D$  on  $N_W$ , the device width is varied from a minimum of 10 nm–50 nm. Moreover, the  $N_T$  is also varied from 5 nm to 9 nm to see the impact of  $N_T$  on  $I_D$ . Fig. 5(a) depicts the ON current ( $I_{ON}$ ) of the device at  $V_{GS} = V_{DS} = 0.7 \text{ V}$ , which is a crucial measure for high performance applications. Also, the proportionality increment in  $I_{ON}$  with respect to effective width gives rise to higher currents when the

nanosheet's geometry increases. The highest increment of 286% in  $I_{ON}$  is observed for  $N_T$  of 5 nm for the  $N_W$  ranging from 10 nm to 50 nm.

The OFF current ( $I_{OFF}$ ) variations are shown in Fig. 5(b) with respect to various width and thickness values of the nanosheet. In addition to the increment in  $I_{ON}$ ,  $I_{OFF}$  also gets incremented with the increment in the geometry of NS and is shown in Fig. 5(b). This phenomenon is observed because, increment in  $N_W$  leads to more leakages in the device because of the degraded control of gate over the channel. It is evident that by using thinner nanosheets, the  $I_{OFF}$  is mitigated. In the OFF state, as the  $N_T$  increases, the potential barrier height and conduction band



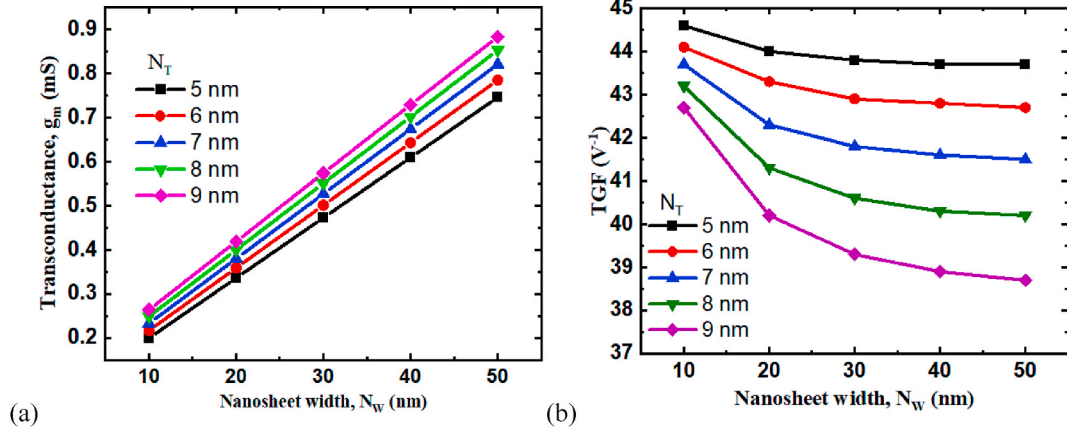


Fig. 8. (a) Transconductance ( $g_m$ ) and (b) Transconductance generation factor (TGF) values as a function of  $N_w$  for various  $N_t$  values.

energy in the channel gets reduced and leads to an increment in leakage currents. However, as the thickness decreases, the  $I_{ON}$  decreases marginally due to mobility degradation caused by increased perpendicular electric field [33]. An increment of 2 orders in  $I_{OFF}$  is observed with the increment in  $N_w$  from 10 nm to 50 nm at  $N_t$  of 9 nm. As  $N_t$  increases, a large increment in the  $I_{OFF}$  is observed due to the degradation of electrostatic integrity over the channel.

Fig. 6(a) depicts  $I_{ON}/I_{OFF}$  variations with respect to  $N_w$  for various values of  $N_t$ . For better logic applications, the switching ratio of  $10^6$  is feasible. It is observed from Fig. 6(a) that the switching ratio performance deteriorates as the  $N_w$  increases due to large increment in  $I_{OFF}$ . A large increment in the  $I_{OFF}$  with a marginal increment in  $I_{ON}$  tends to downfall in the  $I_{ON}/I_{OFF}$  as  $N_w$  increases. Moreover, the switching ratio decreases as the  $N_t$  increases, which is because of a huge increment in  $I_{OFF}$  due to reduced electrostatic integrity of the gate over the channel.

The highest decrement of 89% in switching ratio is noticed for  $N_t$  of 9 nm as the  $N_w$  towards 50 nm. Degradation of switching ratio is more as the  $N_t$  increases due to weak electrostatic control of gate over the channel.

Threshold voltage ( $V_{th}$ ) variations are shown in Fig. 6(b). The  $V_{th}$  gets decrement as the  $N_w$  increases due to an increment in leakage currents. The same trend is observed as the  $N_t$  is increased. It is seen that a downfall of 2.9% and 5.4% in  $V_{th}$  for  $N_w$  of 10 nm and 50 nm, as the  $N_t$  ranges from 5 to 9 nm. Similarly, a fall of 5.3% and 7.6% in  $V_{th}$  are observed for the  $N_t$  of 5 and 9 nm, as the  $N_w$  is increased. As the  $N_t$  and  $N_w$  values are increasing, the  $V_{th}$  roll-off is more since the deterioration of gate electrostatics on the channel region.

SS is an essential parameter for the evaluation of subthreshold performance towards low power applications. SS gives the value of  $V_{GS}$  required to get a decade change in  $I_D$ . Lower SS values are preferred for the good subthreshold operation of the device. The SS takes the expression (1).

$$SS = \left( \frac{\partial \log_{10} I_D}{\partial V_{GS}} \right)^{-1} \quad (1)$$

SS values as a function  $N_w$  for various  $N_t$  values are given in Fig. 7 (a). The NSFET with geometrical values of  $N_w$  and  $N_t$  of 10 nm and 5 nm performs best with an SS of 62.5 mV/dec, which is close to the optimum value.

As the  $N_w$  is increasing, a maximum increment of 8.6% in SS is found for  $N_t$  of 9 nm. As the  $N_t$  is raised, a maximum growth of 10.6% in SS is noticed for  $N_w$  of 50 nm. The sub-threshold performance degrades for higher  $N_w$  and  $N_t$  values because of higher SS.

DIBL is another important sub-threshold performance measure, and low value of DIBL is preferred for optimal device performance. Equation (2) is used to calculate the DIBL.

$$DIBL = \frac{V_{thlin} - V_{thsat}}{V_{Dsat} - V_{Dlin}} \quad (2)$$

Here,  $V_{thlin}$  is the  $V_{th}$  taken at  $V_{Dlin} = 0.04$  V and  $V_{thsat}$  is  $V_{th}$  taken at  $V_{Dsat} = 0.7$  V. The  $V_{th}$  is calculated through the constant current method at  $200\text{ nA} \times W_{eff}/L_G$ . The effective width ( $W_{eff}$ ) is the perimeter of the nanosheet [32,33].

Fig. 7(b) depicts DIBL values as a function of  $N_w$  for various  $N_t$  values. It can be seen that when the dimensions of the nanosheets grow, the DIBL grows as well. By integrating nanosheets with lesser width and thickness, the DIBL can be reduced due to low  $V_{th}$  roll-off. For  $N_w$  50 nm, the highest increment of 30.6% in DIBL is observed. Also, for  $N_t$  of 9 nm, the highest increment of 62% in DIBL is noticed. From this, it can be inferred that DIBL is sensitive to higher width and thickness variations.

### 3.2. Geometry influence on Analog/RF response of GAA NSFET

The aggressive scaling of devices made it possible to build system-on-chip (SOC) devices. Analog and RF circuits are embedded on SOC for various applications. Thus, the study of analog and RF metrics of a device is very crucial to estimate the potential of the device towards analog and RF applications. In this section, various analog/RF metrics like  $g_m$ , TGF,  $g_{ds}$ , intrinsic gain, cut-off frequency, intrinsic delay, TFP, GFP, GTFP, and GBW are studied for different geometries of NSFET.

Transconductance ( $g_m$ ) is one of the essential analog metrics which evaluates the change in  $I_D$  obtained to the corresponding change in  $V_{GS}$ . Inherently  $g_m$  gives the speed of a device, and better  $g_m$  indicates that the device can operate at high speed for logic operations [36]. Moreover, the gate transport efficiency is high for devices which are having high  $g_m$ . Transconductance decides the amplifying capability of the device. A higher amount of gain or amplification can be obtained by having more  $g_m$  values for analog applications.  $g_m$  can be calculated by using equation (3).

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \quad (3)$$

Fig. 8(a) depicts the  $g_m$  variations for various  $N_w$  and  $N_t$  values. It is noticed that for the lesser nanosheet widths, the value of  $g_m$  is less. Higher  $g_m$  values can be obtained by increasing the nanosheet thickness and width. Due to the peak electron velocity at the source region and less channel doping in the channel region, higher  $g_m$  is obtained. As the  $N_w$  increasing, the highest increment of 273% in  $g_m$  is observed for  $N_t$  of 5 nm. Also, the highest increment of 32% in  $g_m$  is observed for  $N_w$  of 10 nm as the  $N_t$  increases towards upper bound. It is noticed that the improvement in  $g_m$  is due to the rise in drain current for larger  $N_t$  and  $N_w$  values.

TGF =  $g_m/I_D$ , is another crucial analog FOM and it estimates the

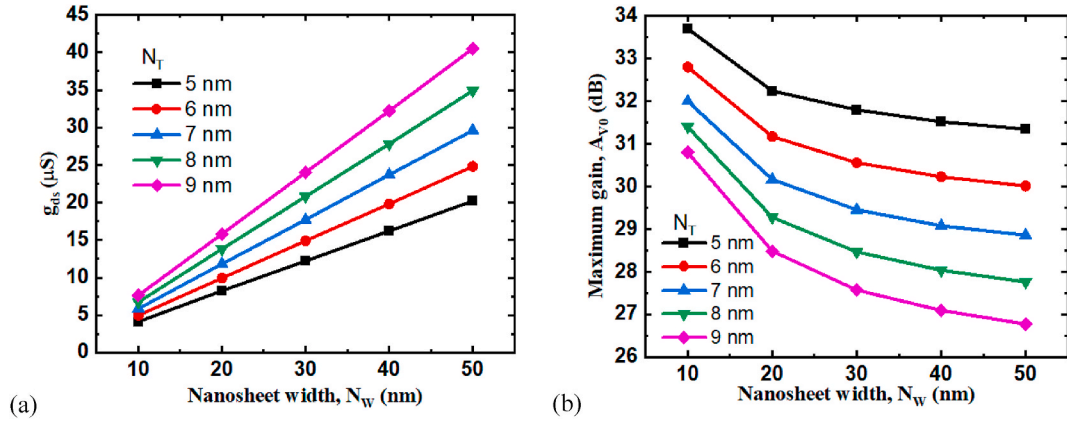


Fig. 9. (a)  $g_{ds}$  and (b) Maximum gain,  $A_{V0}$  values as a function of  $N_W$  for various  $N_T$  values.

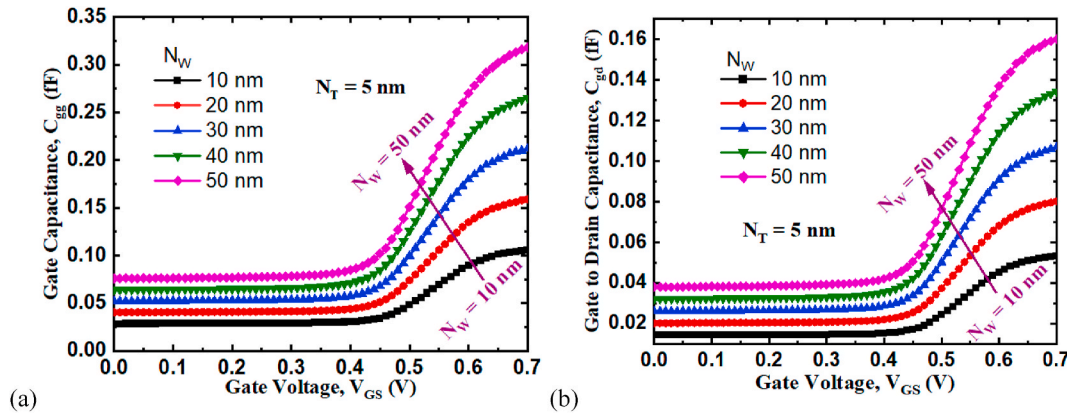


Fig. 10. (a)  $C_{gg}$  and (b)  $C_{gd}$  versus  $V_{GS}$  for various  $N_W$  values for constant  $N_T$  of 5 nm.

efficient use of drain current to obtain the agreeable value of  $g_m$ . It is also taken as accessible gain per unit power dissipation. The device with more TGF values can be operated at lower voltages without deteriorating the performance [37]. Also, TGF is used to estimate the power required to achieve high speed [38]. The variations in TGF as a function  $N_W$  for various  $N_T$  values are depicted in Fig. 8(b). As the  $N_T$  and  $N_W$  values are increased, a reduction in TGF is noticed. In the weak inversion region, higher TGF values are obtained and tend to decrease in the super threshold region due to velocity saturation. In addition to the increase in  $g_m$ , an increase in  $I_D$  is noticed when the  $N_W$  and  $N_T$  increase due to the rise in effective width. There is a growth rate of 273% in  $g_m$  observed when the  $N_W$  is increased from 10 nm to 50 nm. However, the growth rate of 286% in  $I_D$  is observed when the width is increased from 10 nm to 50 nm, which is more than the growth rate of  $g_m$ . This makes the TGF downfall in spite of the increment in  $g_m$ . The highest 8.8% in TGF is observed for  $N_T$  of 9 nm as the  $N_W$  is increased from 10 nm to 50 nm. Also, the highest decrement 8.3% in TGF is observed for  $N_W$  of 50 nm as the  $N_T$  is raised from 5 to 9 nm. It can be seen that the deterioration in TGF is observed for increment in  $N_W$  and  $N_T$  values of NSFET.

Another essential analog/RF figure of merit is output conductance ( $g_{ds}$ ) and it should be minimum for good performance of the device. Mostly, the devices incorporated in analog circuit design are operated at the saturation region. Ideally, in the saturation region, the drain current is independent of drain voltage. However, because of short channel effects like DIBL, the variation in drain voltage leads to variations in drain current also. To measure the dependency of drain current on drain voltage in the saturation region, the parameter  $g_{ds}$  is evaluated, and for better performance, the  $g_{ds}$  values should be as minimum as possible. The  $g_{ds}$  is given by equation (4).

$$g_{ds} = \frac{\partial I_D}{\partial V_{DS}} \quad (4)$$

Fig. 9(a) depicts  $g_{ds}$  variations for various nanosheet width and thickness values. It is clearly seen that for nanosheet width of 10 nm and thickness of 5 nm outperforms remaining all the variations, and it can be concluded that for higher values of nanosheet's width and thickness, the amplification capacity of the device degrades since  $g_{ds}$  is increasing. Moreover, for thicker nanosheets,  $g_{ds}$  is increasing drastically. The highest increment of 431% in  $g_{ds}$  is noticed for  $N_T$  9 nm respectively as the  $N_W$  is increased from 10 nm to 50 nm. Also, an increment of 101% in  $g_{ds}$  is observed for  $N_W$  of 50 nm as the  $N_T$  is increased from 5 to 9 nm.

Intrinsic gain for various  $N_W$  and  $N_T$  variations is shown in Fig. 9(b). It is noticed that with the increment in  $N_W$ , the gain is deteriorating because of the increased dependency of drain current on drain voltage. Similarly, the higher gain is obtained for thinner nanosheets because of the decrement in output conductance and enhanced gate electrostatics over the channel. The highest decrement of 13.1% in gain is observed for  $N_T$  of 9 nm as the  $N_W$  is increased from 10 nm to 50 nm. Also, the highest deterioration of 14.6% in gain is observed for  $N_W$  of 50 nm as the  $N_T$  is increased from 5 to 9 nm. For higher values of  $N_W$  and  $N_T$  the gain is deteriorating because of the increment in output conductance.

Fig. 10(a) and (b) show the  $C_{gg}$  and  $C_{gd}$  variations as a function of  $V_{GS}$  for various  $N_W$  values at a constant  $N_T$  of 5 nm. Also, Fig. 11(a) and (b) depict the  $C_{gg}$  and  $C_{gd}$  variations with  $V_{GS}$  for various  $N_T$  values of the nanosheet at a constant  $N_W$  of 10 nm. The gate capacitance is the summation of both  $C_{gs}$  and  $C_{gd}$ , i.e.,  $C_{gg} = C_{gd} + C_{gs}$ . It can be deduced from Figs. 10 and 11 that as the nanosheet dimensions grow, the  $C_{gg}$  and  $C_{gd}$  increase due to the increase in effective area.

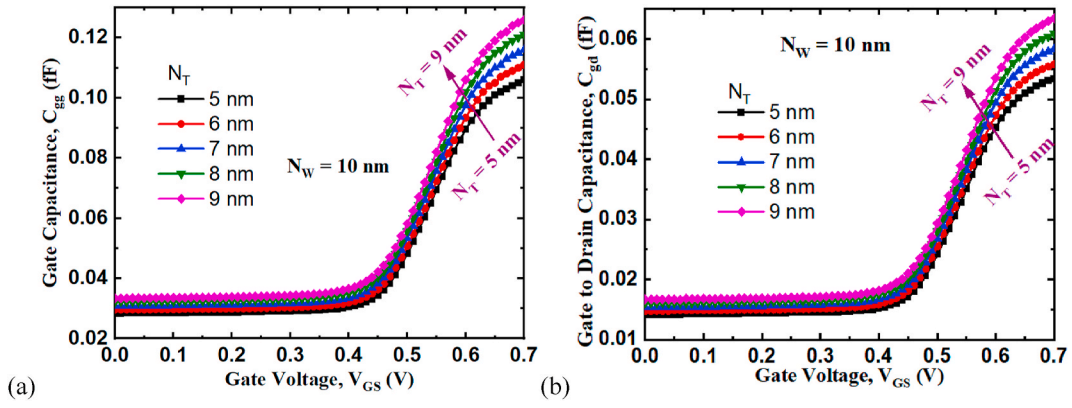


Fig. 11. (a)  $C_{gg}$  and (b)  $C_{gd}$  values as a function of  $V_{GS}$  for various  $N_T$  values for constant  $N_W$  of 10 nm.

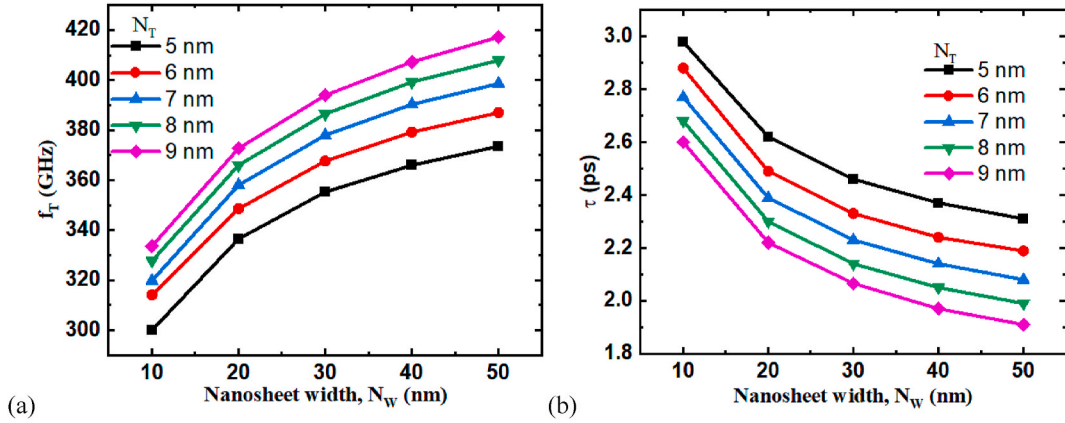


Fig. 12. (a)  $f_T$  (b) Intrinsic delay,  $\tau$  values with respect to  $N_W$  for different  $N_T$  values.

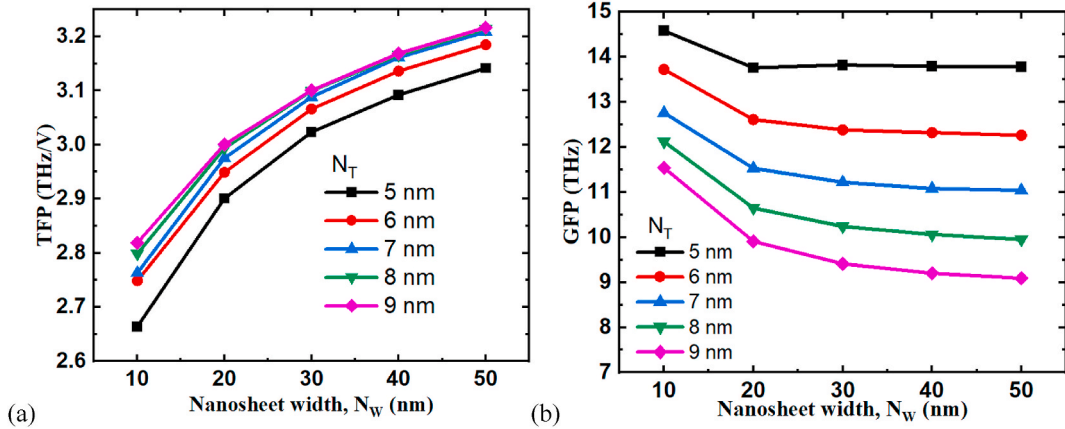


Fig. 13. (a) TFP and (b) GFP values as a function of  $N_W$  for various  $N_T$  values.

To analyze a device for its suitability for high frequency applications, it is crucial to consider the Miller capacitance ( $C_{gd}$ ) [39]. Miller capacitances are capacitances between the output and input that mitigate the device's gain performance at high frequencies. The crucial FOMs, which are dependent on Miller capacitances, are discussed further in this paper.

Cut-off frequency ( $f_T$ ) is another important characteristic to consider when evaluating the device's analog/RF performance. Usually, the  $f_T$  is derived by taking into account of the frequency at which the magnitude of current gain is equal to one. The cut-off frequency is defined as [40].

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})} \quad (5)$$

From Fig. 12(a), the variations in  $f_T$  are observed from different geometrical variations of NSFET. It can be seen that with the raise in  $N_W$ , the  $f_T$  followed an increasing manner. As the  $N_W$  and  $N_T$  increase, the transconductance, effective width and capacitance values increase and cause an increase in cut-off frequency. The maximum increment of 24.6%, in cut-off frequency, is obtained for  $N_T$  of 7 nm  $N_W$  increases from lower to upper bound. Also, the highest increment of 10.8% in  $f_T$  is observed for  $N_W$  of 20 nm as  $N_T$  is increased from lower to upper bound.

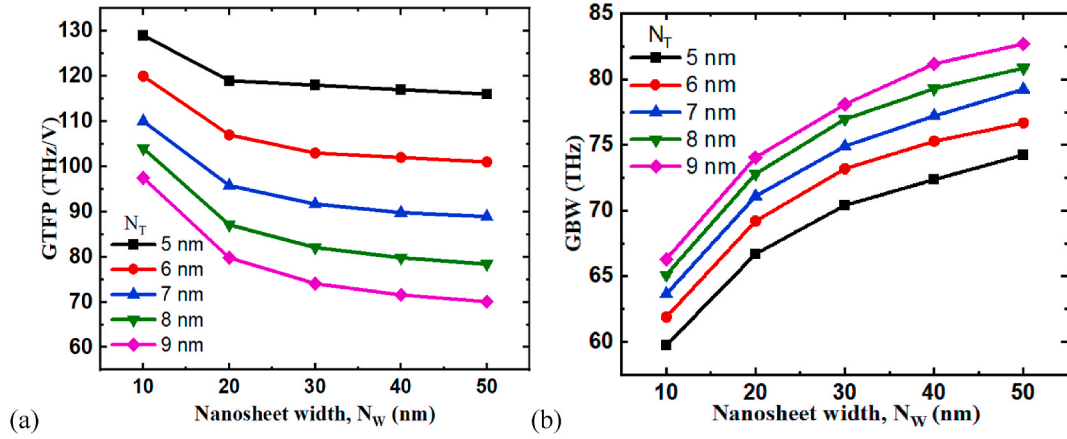


Fig. 14. (a) GTFP and (b) GBW values as a function of  $N_W$  for various  $N_T$  values.

Table 2

Comparison of DC and analog/RF metrics with existing literature.

| Reference | $I_{ON}$ (A)                     | $I_{OFF}$ (A)                     | $I_{ON}/I_{OFF}$   | SS (mV/dec) | DIBL (mV/V) | $g_m$ (S)            | $f_T$ (GHz) |
|-----------|----------------------------------|-----------------------------------|--------------------|-------------|-------------|----------------------|-------------|
| [33]      | $6.8 \times 10^{-5}$ A           | $6.7 \times 10^{-12}$ A           | $2.3 \times 10^7$  | 71          | 22.8        | –                    | 404         |
| [46]      | $16.6 \times 10^{-5}$ A          | –                                 | –                  | –           | 32          | $78 \times 10^{-5}$  | 607         |
| [47]      | $2.37 \times 10^{-5}$ A/ $\mu$ m | $1.15 \times 10^{-10}$ A/ $\mu$ m | $2 \times 10^5$    | 70.89       | –           | $5.4 \times 10^{-5}$ | 3790        |
| [48]      | 832 $\mu$ A/ $\mu$ m             | 13.3 nA/ $\mu$ m                  | $6.3 \times 10^4$  | 74          | –           | –                    | –           |
| This work | $2.45 \times 10^{-5}$ A          | $1.24 \times 10^{-14}$ A          | $1.98 \times 10^9$ | 62.48       | 31.81       | 2E-4                 | 300         |

The intrinsic delay,  $\tau$  is shown in Fig. 12(b). Basically,  $\tau$  can be obtained by taking the product of total resistance and total capacitances [41].  $\tau$  is very important FOM to evaluate the device's practicality in SOC applications [42]. The intrinsic delay is given by the expression (6)

$$\tau = \frac{C_{gg} V_{DD}}{I_{ON}} \quad (6)$$

From equation (6), it can be seen that,  $\tau$  is proportional to the total capacitance  $C_{gg}$  and power supply and is inversely proportional to the  $I_{ON}$  of the device. It is obvious that as the effective width of the nanosheet increases, both on current and gate capacitance of the device are boosted up. As the  $N_W$  is increased from 10 nm to 50 nm, there is an enhancement of 4x in  $I_{ON}$  and 2x in  $C_{gg}$  for a fixed  $N_T$  of 5 nm. So compared to the improvement in on current, the increment in the gate

capacitance is less. Thus, there is a downfall in  $\tau$  as the  $N_W$  ranges from 10 nm to 50 nm. It is noticed that the lowest delay characteristics are obtained for an  $N_W$  of 50 nm. The decrement of 22.5% and 26.5% in  $\tau$  is observed for  $N_T$  values of 5 nm and 9 nm, respectively, as the  $N_W$  raised from 10 nm to 50 nm. Also, the decrement of 12.7% and 17.3% in the  $\tau$  is observed for  $N_W$  of 10 nm and 50 nm, respectively, as the  $N_T$  is increased from 5 to 9 nm.

The TFP is shown in Fig. 13(a). TFP is another important analog/RF figure of merit and is obtained by taking the product of trans-conductance and cut-off frequency, given by equation (7) [43].

$$TFP = \frac{g_m}{I_D} \times f_T \quad (7)$$

TFP gives the trade-off between bandwidth and power and is used in medium and high-speed circuit applications [44]. From Fig. 13(a), it can be observed that TFP is more for higher  $N_T$  and  $N_W$  values because of an increment in cut-off frequency. The highest increment of 17.9% and the lowest increment of 14.11% are observed as the  $N_W$  is varied from 10 nm to 50 nm for  $N_T$  of 5 nm and 9 nm, respectively. Also, the maximum increment of 5.8% and minimum increment of 2.4% in TFP are noticed for  $N_W$  of 10 nm and 50 nm, respectively, as the  $N_T$  is increased from 5 nm to 9 nm.

Gain frequency product (GFP) is another crucial analog/RF figure of merit [45], given by

$$GFP = \frac{g_m}{g_{ds}} \times f_T \quad (8)$$

As the intrinsic gain is the most important parameter for operational amplifiers, GFP is depicted in Fig. 13(b) is used to estimate the performance of operational amplifiers in high frequency circuit implementations. There is a decrement in GFP observed for increment in  $N_W$  and  $N_T$  values. The decrement of 5.5% and of 21% in GFP were observed as the

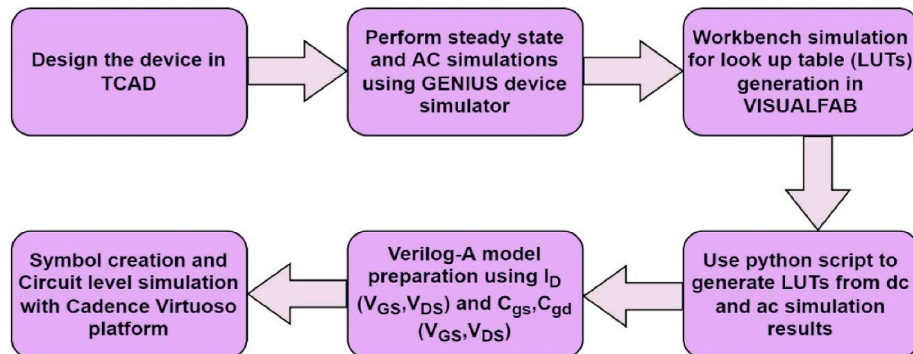


Fig. 15. Flow chart of verilog- A model creation and SPICE simulation.



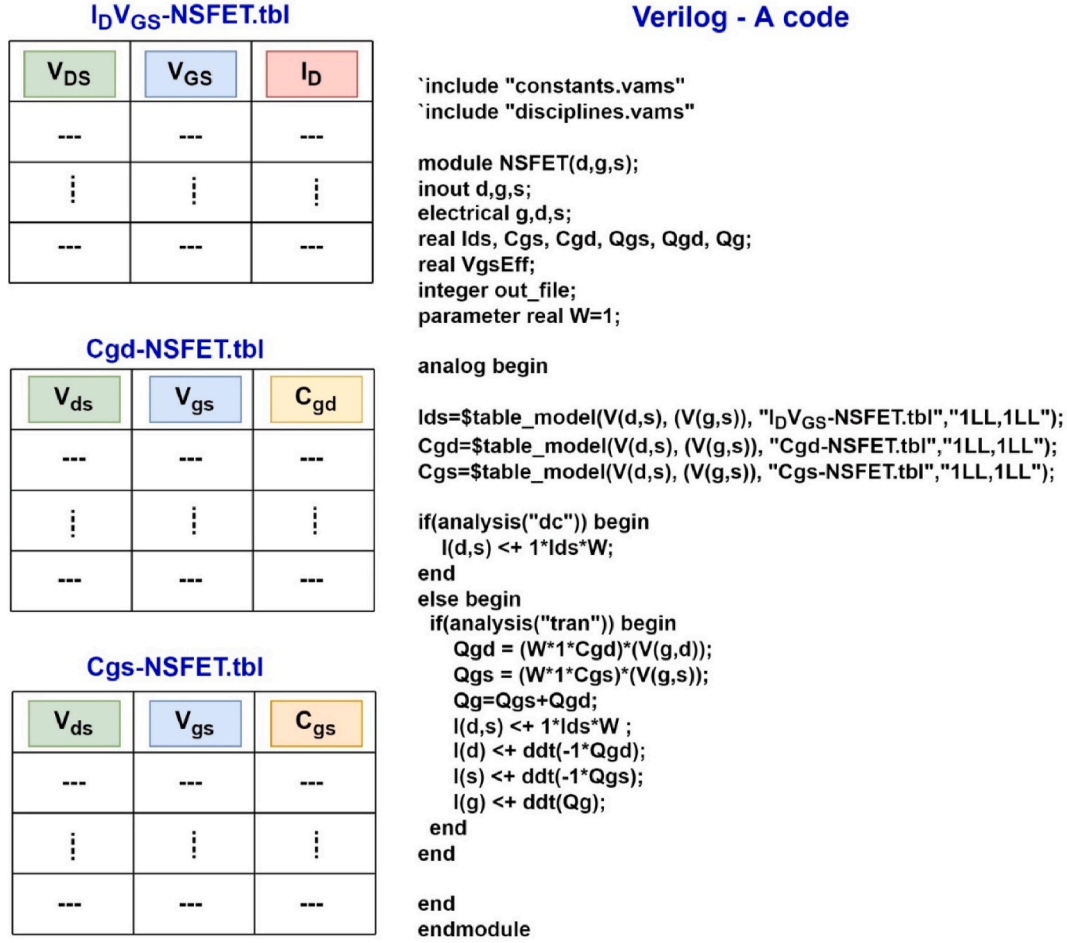
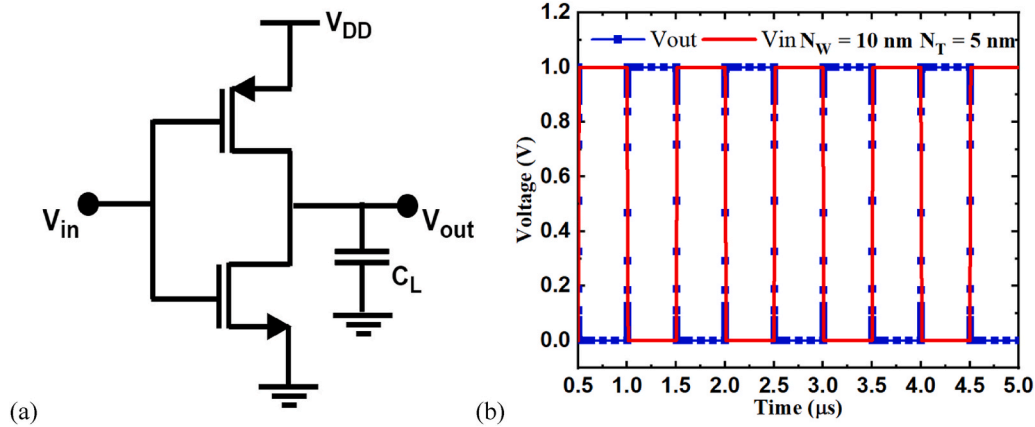


Fig. 16. Look up table format (left) and Verilog-A code for symbol creation (right).

Fig. 17. (a) Inverter schematic diagram (b) transient response of inverter for  $N_W = 10$  nm and  $N_T = 5$  nm.

$N_W$  varies from 10 nm to 50 nm for  $N_T$  of 5 nm and 9 nm, respectively. Also, the decrement of 21% and 34% in GFP are observed for  $N_W$  of 10 nm and 50 nm, respectively, as the  $N_T$  is increased from 5 nm to 9 nm. More deterioration in GFP is observed for higher  $N_T$  and  $N_W$  values since there is an increment in the  $g_{ds}$ .

The gain transconductance frequency product (GTFP) is very much useful for circuit designers. GTFP allows the designers to identify the optimum region achieving a good trade-off among parameters like transconductance, speed, and gain of the device [44]. GTFP is given by (9)

$$GTFP = A_{V0} \times TFP = \frac{g_m}{g_{ds}} \times \frac{g_m}{I_D} \times f_T \quad (9)$$

The GTFP as a function of nanosheet width for various nanosheet thickness values is depicted in Fig. 14(a). It can be seen that by using less  $N_W$  and  $N_T$  for nanosheet, higher GTFP can be achieved. For  $N_T$  of 5 nm and 9 nm, respectively, the decrement of 10% and 28% in GTFP were found when the  $N_W$  varied from 10 nm to 50 nm. Also, the decrement of 24% and of 40% in GTFP is observed for  $N_W$  of 10 nm and 50 nm respectively as the  $N_T$  is increased from 5 nm to 9 nm.

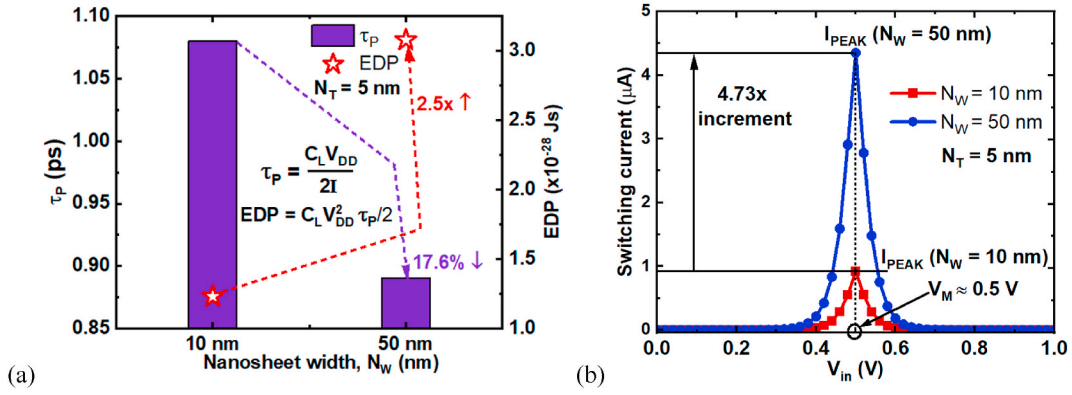


Fig. 18. (a) Propagation delay ( $\tau_p$ ), EDP and (b) Switching current of inverter for different  $N_W$  at constant  $N_T$  of 5 nm.

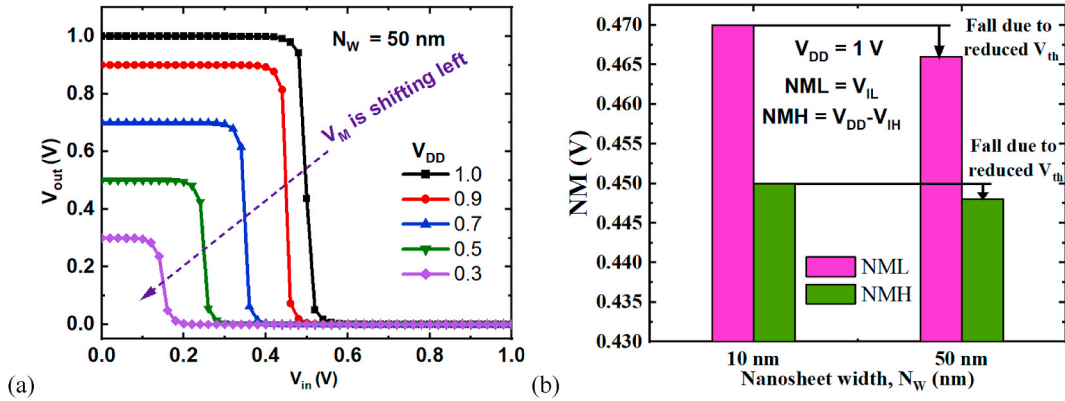


Fig. 19. (a) Transfer characteristics of inverter for various  $V_{DD}$  values (b) NM as a function of  $N_W$  at  $V_{DD} = 1$  V.

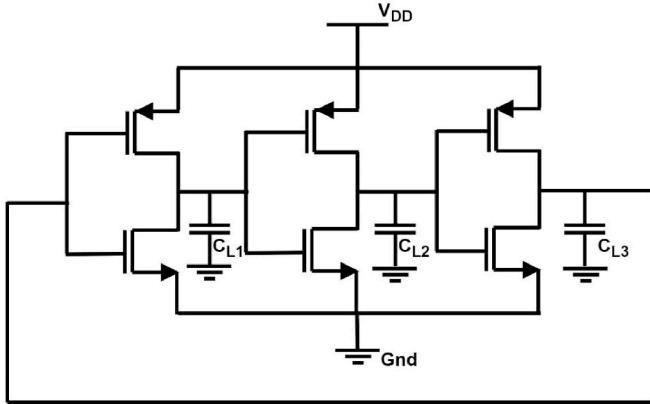


Fig. 20. 3-stage ring oscillator (RO).

Another crucial analog/RF FOM is gain bandwidth product (GBW) and is used to evaluate device efficiency in high frequency applications. GBW can be calculated by the following equation [45].

$$GBW = \frac{g_m}{20\pi C_{gd}} \quad (11)$$

From Fig. 14(b), it can be seen that as the width and thickness increase, the increment in GBW is observed since it is proportional to transconductance ( $g_m$ ). Miller capacitance or gate to drain capacitance ( $C_{gd}$ ) increases with increment in width and thickness values. However, the improvement in GBW is observed because of a large improvement in  $g_m$  compared to  $C_{gd}$ . An increment of 23.8% and 24.7%, in GBW is

observed for thickness values of 6 nm and 9 nm respectively as the  $N_W$  is increased from 10 to 50 nm. Moreover, an increment of 10.93% and 12.2% in GBW is observed for  $N_W$  of 30 nm and 40 nm, respectively, as the  $N_T$  is increased from 5 to 9 nm. The comparison of DC and analog/RF metrics with existing literature is given Table .2.

#### 4. Circuit analysis

In this section, the impact of nanosheet width on inverter and ring oscillator's performance is analysed by performing the SPICE simulation using the look up table-based Verilog-A model [39]. The flow of the SPICE simulation is shown in Fig. 15.

Initially, the device is designed using the Visual TCAD tool [31]. On top of that, the DC and AC characteristics are obtained using the Visual Fab parallel simulation platform. From the obtained characteristics, the look up tables are prepared using the format shown in Fig. 16. By using the look up tables and Verilog -A model, the symbol is created in the Cadence Virtuoso tool [50] and is further used in schematic diagrams. Fig. 17 (a) depicts the schematic of the inverter diagram. In the inverter diagram,  $V_{DD}$  is the supply voltage,  $V_{in}$  and  $V_{out}$  are the input and output voltages, respectively. The transient response for  $N_W$  of 10 nm is shown in Fig. 17(b). For digital applications, it is important to evaluate the parameters like propagation delay ( $\tau_p$ ), energy delay product (EDP), switching current, and noise margins to estimate the digital performance of the device. For the designed inverter, these parameters are calculated and compared the performance variations for  $N_W$  of 10 nm and 50 nm at a constant  $N_T$  of 5 nm.

Various transient and dc responses of the inverter are simulated using the Cadence Virtuoso platform. The load capacitance  $C_L$  is taken as the sum of  $C_{gg}$  of both N-NSFET and P-NSFET [49]. The transient response of the inverter is shown in Fig. 17(b) for the time of 5  $\mu$ s.

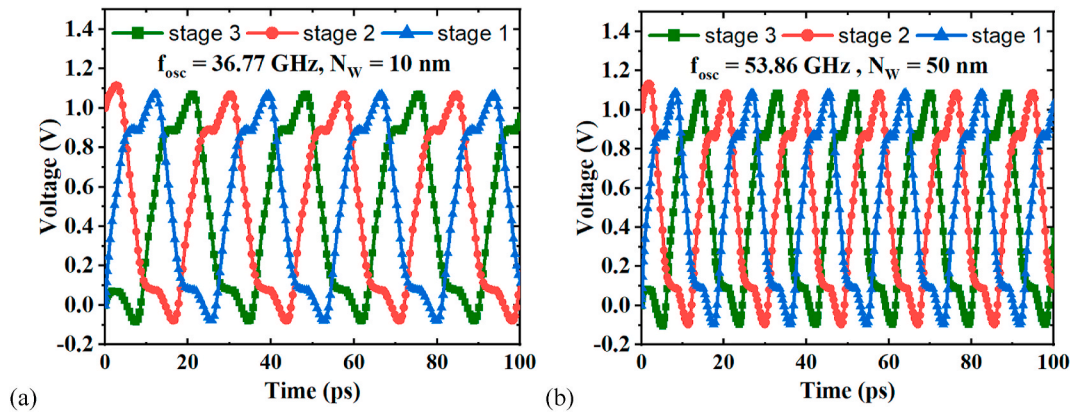


Fig. 21. 3-stage ring oscillator's transient response for (a)  $N_W$  of 10 nm and (b)  $N_W$  of 50 nm.

**Table 3**  
Comparison of  $f_{osc}$  with existing literature.

| Reference                  | Number of stages | $f_{osc}$ (GHz) | Technology node |
|----------------------------|------------------|-----------------|-----------------|
| [54]                       | 3                | 52.44           | 22              |
| [55]                       | 3                | 6.02            | 90              |
| [56]                       | 3                | 41.10           | 30              |
| [57]                       | 3                | 2.45            | 180             |
| [58]                       | 9                | 9               | 50              |
| This work ( $N_W = 10$ nm) | 3                | 36.77           | 5               |
| This work ( $N_W = 50$ nm) | 3                | 53.86           | 5               |

Fig. 18(a) shows the propagation delay of the inverter for  $N_W$  of 10 nm and 50 nm. The overall propagation delay ( $\tau_p$ ) was obtained by considering the formula shown in Fig. 18(a) [51]. Where  $C_L$  is the load capacitance,  $I$  is the average current in the circuit. As  $\tau_p \propto C_L \times (1/I_{ON} (N_{NSFET}) + 1/I_{OFF} (P_{NSFET}))$ , with the increment in  $N_W$ , both  $I$  and  $C_L$  will increase. However, the relative increment in  $I$  is higher than that of  $C_L$  and causes a decrement in  $\tau_p$ . From the result analysis, it is noticed that as the  $N_W$  increases from lower to upper bound, the  $\tau_p$  decreases by 37%. Fig. 18(a) also shows EDP of the inverter for different  $N_W$  values. An increment of 2.5x is noticed in EDP as the  $N_W$  increases towards its upper bound.

Fig. 18(b) depicts the switching current ( $I_{SC}$ ) of inverters for  $N_W$  values 10 nm and 50 nm. The inverter with higher  $N_W$  is delivering more switching current, and an increment of 4.73x is observed in peak switching current with  $N_W$  of 50 nm. Also, the static current (at  $V_{in} = 0$  V,  $V_{in} = V_{DD}$ ) is less than 10 pA and offers much lower static power dissipation and energy efficiency. Moreover, the switching threshold ( $V_M$ ), which is the point where  $V_{in} = V_{out}$  is much closer to the ideal value, which is 0.5 V. Fig. 19(a) shows the DC response for various supply voltages ( $V_{DD}$ ) for  $N_W$  of 50 nm. It is noticed that as the  $V_{DD}$  is decreasing, the  $V_M$  is shifting towards left and affecting the switching characteristics.

Fig. 19(b) shows the noise margins (NM) for different nanosheet widths. The noise margin low (NML) and noise margin high (NMH) are calculated using the formulas shown in Fig. 19(b).  $V_{IL}$  and  $V_{OH}$  are the input voltages where the gain of the inverter ( $dV_{out}/dV_{in} = -1$ ) [52]. It is found that the NM followed the threshold voltage and got decrement with rise in  $N_W$ . Also, the noise margin is higher than its ideal value ( $V_{DD}/2$ ) by 83% for  $N_W$  of 50 nm, which is good for static logic applications [53].

Fig. 20 shows the 3-stage ring oscillator (RO) in which all the load capacitances are assumed to be as same. Initially, an input of 0 V was forced to initiate the oscillations.

From Fig. 21(a) and (b), it is observed that the frequency of oscillations ( $f_{osc}$ ) increases with the width of the nanosheet. As the  $\tau_p$  is decreasing with the increment in  $N_W$ , the  $f_{osc}$  increases with width.

There is an increment of 46.5% observed for  $N_W$  increment from 10 nm to 50 nm due to higher  $I_{ON}$ . As a result of the foregoing research, greater widths assure device flexibility for high-speed applications. The comparison of  $f_{osc}$  with existing literature is shown in Table 3.

## 5. Conclusion

The DC and analog/RF performance of NSFET is investigated in this research by altering the geometry of each nanosheet at the 5 nm technology node. As the  $N_T$  and  $N_W$  are raised, there is an increase in both  $I_{ON}$  and  $I_{OFF}$ . The optimum values for switching ratio, DIBL, and SS are obtained for lower values of  $N_W$  and  $N_T$ . However, analog/RF FOMs like  $g_m$ ,  $g_{ds}$ ,  $f_T$ , TFP, and GBW are more for higher  $N_W$  and  $N_T$  values of NS. These results will give an understanding of DC and analog/RF performance of NSFET with geometrical variations of the device. Moreover, the circuit performance was assessed for circuits like inverter and 3-stage RO and found that with the increment in width, the propagation delay gets decremented, and  $f_{osc}$  increases. These results will give performance insights of GAA NSFET at both device and circuit levels.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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